

Title

Digital health intervention for childhood weight management in preschools: outcomes at nine months from a cluster randomized trial

Abstract

Background: The efficacy of digital health tools in paediatric obesity prevention remains incompletely established, particularly in lower-middle income settings. **Methods:** We randomized 14 kindergartens (980 children, mean age 5.1 years) to receive a 9-month digital health platform intervention (interactive health tracking, dietary guidance, family engagement modules) or standard programming. The primary outcome measured between-group changes in BMI z-scores at month 9. **Results:** In the intervention group, mean BMI z-score changed by -0.18 (95% CI -0.31 to -0.05), compared to -0.06 (95% CI -0.19 to 0.07) in control sites (between-group difference -0.12; $p=0.034$). Dietary quality scores improved in both groups, with slightly greater gains in the intervention arm. Physical activity levels were similar between groups. No adverse events were reported. **Conclusion:** The digital health intervention produced modest but statistically significant reductions in relative weight gain in early childhood, with feasibility demonstrated for school-based deployment.